

DRILLTICK

App Overview

A drilling-operations and safety toolkit for the rig floor and the office

⚠ FOR UNDERSTANDING PURPOSES ONLY

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Flutter / Dart

Well Control

MWD tools

Safety & permits

Lifting

1. What DrillTick Is

DrillTick is a mobile and desktop toolkit for people who work on drilling rigs. It brings together the calculators, simulators, forms and reference guides that a drilling or safety professional uses every day, so that a kill sheet, a work permit, a safety checklist or a crane lift plan can be filled in on a phone or tablet, saved, and exported as a clean PDF.

Everything is stored **on the device itself** in a local database. There are no accounts and no servers. The app is organised into two main areas, **Operations** and **Safety**, joined by a simple three-button bottom bar.

The idea in one line

The paper forms, charts and rules of thumb of a drilling site — in one dark-themed, rig-friendly app that calculates, checks and exports.

2. At a Glance

2

main areas

4

well-control tools

7

checklists

6

work permits

46

source files

Type	Personal learning project, designed, built and documented by one developer
Built with	Flutter and Dart
Look and feel	Dark theme with a deep navy background and a vivid green accent; each module has its own accent colour
Where data lives	On the device, in a local database. No cloud and no login
Output	A4 PDFs for sheets, permits, checklists and reports; Excel and CSV exports

Size	About 46 source files and roughly 181,000 lines of Dart code
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3. The Map of the App

Home has four buttons: **DrillTick Operations**, **DrillTick Safety**, **Contact Us** and **Quit**. Operations and Safety each open a tree of tiles.

Area	Built	Shown as “coming soon”
Operations › Drilling › Well Control	Surface and Subsea Kill Sheets, Surface and Subsea SCR Measure	Shut-in procedures, kick detection, well-control methods
Operations › LWD / MWD › MWD	Rig Sensor, Survey and Downlinking simulations	LWD, Directional Drilling, Start a New Job, MPT
Safety › Report An Event	Incident, Near Miss and three Observation forms	—
Safety › Safety Dashboard	Seven pages of live charts and an event register	—
Safety › HSE Periodic Reporting	Monthly Report	Daily and Weekly reports
Safety › Checklists	Seven checklists	—
Safety › Process Safety	Permit to Work, Well Control, Lifting Operations	Risk Assessment Report, HAZOP Report
Operations	Drilling and LWD / MWD	Reservoir, Well Testing, EOR

4. Four Themes

Theme	What it contains
Well control	IWCF-style kill sheets for surface and subsea wells, with live calculations and a closely matching PDF; slow-rate pressure (SCR) recording for both
MWD training tools	Hands-on simulations of taking a survey, downlinking to a tool and calibrating rig sensors
Safety management	Event reporting, a performance dashboard, a monthly HSE report, seven checklists and six permit-to-work forms
Lifting operations	A crane lifting plan with load chart, daily check and tandem lifts; a visual lifting manual; and a rigging register with inspections and sling calculations

5. Design Principles

- **One consistent screen pattern.** A header with a back button, a breadcrumb line, collapsible numbered sections and the shared bottom bar.
- **Colour with meaning.** Cyan for data and measurement, red for critical and well control, amber for drill-floor operations, green as the main accent, slate grey for “coming soon”.
- **Calculate as you type.** Results appear immediately and are marked as read-only, so nobody overwrites a calculated value by accident.
- **Save slots and PDFs.** Most forms can be saved into numbered slots and exported as a shareable A4 PDF.
- **Safe rounding.** Where a number sets a limit, it is always rounded in the safe direction.
- **Honest about unfinished work.** Planned modules are visible as “coming soon” tiles.

Important note

DrillTick is a learning and reference tool. It is not a substitute for approved company procedures, certified training or the judgement of competent personnel, particularly for well control, permits to work and lifting operations.

6. Guide to This Document Set

#	Document	Covers
01	Setup & Startup	Packages, theme, splash and the shared bottom bar
02	Home & Navigation	The hubs and the colour language
03	Data Storage & Saved Sheets	The local database and save slots
04	MWD Survey Simulation	Taking a survey on a simulated rig floor
05	MWD Downlinking Simulation	Three ways of sending commands to a tool
06	Rig Drill Sensors	Wiring and calibrating the rig sensors
07	Well Control Hub	The map of well-control tools
08	SCR Measure — Surface	Slow-rate pressures for surface BOP wells
09	SCR Measure — Subsea	The deepwater version with riser margin
10	Kill Sheet — Surface	The eight-calculation IWCF kill sheet
11	Kill Sheet — Subsea	The nine-calculation version with IDCP
12	Safety Hub & Dashboard	The safety menu and seven dashboard pages
13	Report An Event	The entry point and the shared form pattern
14	Incident Report	The most detailed reporting form

#	Document	Covers
15	Near Miss Report	Capturing the event that almost caused harm
16	Observations	Safe behaviour, unsafe act and unsafe condition
17	HSE Monthly Report	Monthly statistics, training and inspections
18	Checklists — Part 1	Orientation, Pre-Spud, Drops and SIMOPS
19	Checklists — Part 2	Inspection / Audit, Mast & Substructure, Contractor
20	Permit to Work	The six permits and the shared five-section form
21	Hot & Cold Work Permits	Work with and without ignition risk
22	Confined Space & Working at Height	Enclosed spaces and fall risk
23	Electrical, Excavation & Radiography	Lock-out, ground work, and a placeholder
24	Crane Lifting Plan	Load chart, daily check, lift plan and tandem lift
25	Lifting Manual	A visual reference for rigging and crane lifts
26	Rigging & Slings	Equipment register, inspection and calculations